Supplemental Material: Shape versus circuit: graphene geometric diodes and Tesla valves as terahertz rectifiers

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This supplement lists the acceptance tests that each solver passed before it was used, the closing convergence checks, and the numerical parameters of the production runs. Every test is a program in the public repository and prints the measured number against its target; the test names below are those of the code, with the plasmonic tests prefixed P to distinguish them from the ballistic ones. Tests T22 to T30 belong to the particle simulation with a computed field, which is used for the 0.55 μm funnel only.

S1. BALLISTIC SOLVER

TABLE S1. Acceptance tests of the trajectory solver at dc (200 000 particles per run). All pass.

Test	What is checked	Target
T1	specular straight channel, no field	transmission exactly 1
T2	diffuse-wall channel, length to width 0.5 to 16	two-dimensional free-molecular transmission factors [1] 0.804, 0.683, 0.541, 0.400, 0.274, 0.179
T3	asymmetric funnel, no field, $p = 1$ and 0.3, $\lambda = 700$ nm	$W_S T_{SD} = W_D T_{DS}$ within 3.5 standard errors
T4	the same funnel with the field on, both injection sides at one total energy	$N_S T_{SD} = N_D T_{DS}$
T5	mirror-symmetric constriction under bias	asymmetry 1
T6	zero bias	zero injection weight and zero current
T7	three-stage side-loop valve, diffuse walls	trapped, leaked and Dirac-cut fraction below 10^{-3}
T8	ballistic 1 μm channel at 10^{12} cm^{-2}	Sharvin conductance 8.74 mS within 1 %
T9	10 μm channel, $\lambda = 300$ nm	$T = \ell_1/(L + \ell_1)$, $\ell_1 = \pi\lambda/2$, within 5 %
T10	chord tolerance 0.05 against 0.005 nm	same transmission
T11	funnel of 100/10/40 nm (lead, neck, drain) at zero field	$T = 0.0995$ against the neck-to-lead ratio 0.100 (information)

TABLE S2. Acceptance tests of the time-dependent solver and the two-sided estimator (10^6 particles per side; T16 4×10^6, T17 2×10^6). Currents in μA for the clean specular funnel at 0.1 V. All pass.

Test	What is checked	Result
T12	integration of $\hbar\dot{k} = F$ without energy re-imposition, static field, against the dc solver	$T(+0.1) = 0.14141$ (reference 0.14130), $T(-0.1) = 0.04148$, $A = 1.622$ (reference 1.624)
T13	two-sided equilibrium estimator against the bias-window estimator; zero bias	188.7 ± 0.6 against 189.2 ± 0.5; -0.15 ± 0.66 at $V = 0$
T14	quasi-static anchor at 300 MHz: rectified current and $\lvert Y_1 \rvert$ against the static curve	17.78 ± 0.59 against 18.27 ± 0.18; 1.589 ± 0.013 against 1.580 mS
T15	straight control at 3 GHz and 1 THz	-0.80 ± 0.57 and -0.20 ± 0.62; trapped plus leaked below 2×10^{-6}
T16	chord cap $v_F/40f$ against $v_F/160f$, 500 nm drop, 1 THz	6.32 ± 0.30 against 5.55 ± 0.32
T17	Dirac cut against a classical turning point, 500 nm drop, 1 THz	5.94 ± 0.45 against 5.69 ± 0.45

The quasi-static anchor is at 300 MHz and not at a few gigahertz because of the slow-drift population described in the main text. The equilibrium injection of the two-sided estimator samples the kinetic distribution down to zero energy, and between 10^{-3} and 7×10^{-3} of the injected electrons are driven below the 2 meV cut that stands in for the Dirac point; these would return to their own contact and carry no current. In the harmonic estimator a drain-injected electron removed by the cut must be booked as returned, otherwise $\lvert Y_1 \rvert$ at the anchor is 6% low. Five operating points

recomputed with a classical turning point in place of the cut agree with the production values within one standard error.

TABLE S3. Acceptance tests of the shape and scale options used for the length scaling (10^6 particles per side). G3 is the 0.55 μm funnel of the main text. Currents in μA at 0.1 V. All pass.

Test	What is checked	Result
T18a	scale invariance, static: device D against twice the current of its half-scale copy, ±0.1 V	188.0 against 188.6 and −117.4 against −117.6; $A = 1.602$ against 1.603
T18b	the same under ac drive: 10 GHz against 20 GHz, 300 GHz against 600 GHz	11.9 ± 0.4 against 12.1 ± 0.5; 0.3 ± 0.5 against -0.4 ± 0.4; admittances equal within two standard errors
T19	scale invariance with scattering: $\lambda = 700$ nm against 350 nm at half scale	static 108.3 ± 0.6 against 110.0 ± 0.5 (2.3 standard errors); ac 2.27 ± 0.23 against 1.82 ± 0.23
T20	mirror-symmetric controls of G2, G3 and G4 at 0.1, 1 and 10 THz	zero; worst case 1.3 standard errors
T21	chord cap $v_F/40f$ against $v_F/160f$, G3, drop at the neck, $\lambda = 700$ nm, 3 THz	3.13 ± 0.10 against 3.20 ± 0.10

Test T21 was first run at 30 THz, where it passed trivially because the response is zero there; the value in the table is from the repeat at 3 THz. In the production runs of the mirror-symmetric controls all 22 points are zero within 2.1 standard errors.

S2. BALLISTIC SOLVER WITH A COMPUTED FIELD

TABLE S4. Acceptance tests of the particle simulation with the computed field. G3 is the 0.55 μm funnel; currents in μA at an amplitude of 0.1 V. All pass, T24 and T26 in the modified form described in the text.

Test	What is checked	Result
T22	interaction kernel: a point charge with its image in the gate; a charged disc over the gate	closed forms to 3×10^{-12} and 1.4×10^{-7}
T23	contacts as equipotentials: capacitance of a plate over the gate, and of two coplanar strips without a gate	0.970 of the plate formula with fringing; 0.990 of the elliptic-integral value
T24	static screening of a weak sinusoidal potential in a uniform sheet	kernel 1.000 ± 0.001; density response of the particles 0.980 of the density of states; loop at weak coupling 1.005 and 0.992 of $1/(1 + \nu G)$; at production coupling the total potential is 0.96 to 0.99 of theory at 1.0 to 2.5 ps and 0.93 to 0.96 at 3.5 to 5.0 ps
T25	speed of the gated plasma wave in a wide strip	0.983 to 0.992 of the collisionless value
T26	interaction off, imposed field: G3 against the trajectory solver, static at ±0.1 V and at 1 THz, both profiles	four of six values within 1.3 standard errors, two static values at 2.0 (0.2 to 0.3% of the current); 1 THz: 2.92 ± 0.05 against 3.03 ± 0.10 and 2.63 ± 0.04 against 2.78 ± 0.11
T27	linear conductance: a straight strip against the Sharvin value; G3 with the interaction on against off	1.006 and 1.002; 1.0004 and 1.0076
T28	conservation: total charge; contact potentials; absorbed power from $\langle VI \rangle$ against the first harmonic	4×10^{-13} electrons; 6×10^{-16} eV; agreement to 0.01%
T29	null tests: no drive, both gates, 20 ps; mirror-symmetric G3 under the shape-only drive at 1 THz	$+0.03 \pm 0.04$, kinetic energy drift 0.04% and 0.00%; -0.08 ± 0.07
T30	convergence of the rectified current of G3 at 1 THz	converged from 4×10^6 particles (Table S5)

The screening test T24 could not be met at 3% at the coupling of the production runs: the total potential is the difference of an applied and an induced potential that cancel to one part in 14 to 30, which amplifies a 0.2% interpolation loss and the relaxation described below. The test was therefore split into the kernel, the particles alone, the closed loop at weak coupling (each to 3%) and the loop at production coupling in an early window (to 10%). T26 was required to pass at two standard errors and passes at 2.5.

The sample particles interact through the grid at short range as well as through the mean field, and these collisions relax the Fermi distribution towards a classical one. In a closed box the static density response is 30% too large after 5 ps at 1.4 particles per nm^2 and 4 to 7% too large at 19 per nm^2, the density of a run of G3 with 4×10^6 particles.

In the open device a particle stays about a picosecond before a contact replaces it, and the rectified current does not change from 4 to 16 $\times 10^6$ particles (Table S5); it is this relaxation, not numerical heating, that sets the smallest usable particle number. Below 4×10^6 the scatter between seeds also exceeds the batch errors.

TABLE S5. Rectified current of the 0.55 μm funnel (μA) at $\lambda = 700$ nm, 1 THz and 0.1 V under the shape-only drive, against the numerical settings. Entries separated by commas are independent seeds. The production setting is 8×10^6 particles, a 5 nm grid, a time step of 2.5 fs, a Gaussian low-pass of 1.5 cells on the kernel and contacts 200 nm long; the variations in the lower part are at 4×10^6 particles, with errors of 0.04 to 0.05.

Setting	Gate at 300 nm	Gate at 30 nm
1×10^6 particles	0.60 ± 0.07	0.55 ± 0.04
2×10^6	0.42 ± 0.03, 0.61 ± 0.06	0.63 ± 0.03, 0.45 ± 0.06
4×10^6	0.47, 0.41, 0.49 ($\pm$0.04)	0.46, 0.45, 0.52 ($\pm$0.05)
8×10^6	0.42 ± 0.03, 0.40 ± 0.02	0.49 ± 0.02, 0.52 ± 0.02
16×10^6	0.42 ± 0.02, 0.44 ± 0.02	0.46 ± 0.01
grid 2.5 nm	0.38	0.52
time step 1.25 fs	0.44	0.48
low-pass 0.75 cells	0.46	0.43
low-pass 3 cells	0.58	0.62
contacts 100 nm long	0.44	0.49
contacts 400 nm long	0.47	0.53
all runs with the production settings and 4×10^6 particles or more	0.429 ± 0.012 (7 runs)	0.483 ± 0.013 (6 runs)

A low-pass of three cells leaves the current 30% too large: the field has to be resolved to 7.5 nm or better. The mirror-symmetric funnel under the shape-only drive, which must give no rectified current, gives -0.003 ± 0.017, $+0.056 \pm 0.028$ and $+0.025 \pm 0.028$ μA at 0.3, 1 and 3 THz under the 300 nm gate and -0.051 ± 0.028, $+0.038 \pm 0.022$ and -0.093 ± 0.052 μA under the 30 nm gate; the weighted mean is $+0.008 \pm 0.023$ μA, and χ^2 against zero is 14.4 for the six runs, so the batch errors are too small by about 1.5 or there is a floor of about 0.05 μA, which is what the main text assumes. With half-diffuse walls the same control gives $+0.018 \pm 0.017$ and -0.052 ± 0.025 μA at 1 THz. Two runs at 3 THz under the 30 nm gate with the drain tied to the gate, whose source-plane and drain-plane currents differed by 0.19 and 0.12 μA after a settling time of 3 ps, were repeated with 10 ps: the strip gives -1.794 ± 0.024 against -1.770 ± 0.026 μA and the funnel $+0.812 \pm 0.026$ against $+0.780 \pm 0.033$ μA, with the two planes then in agreement.

S3. VISCOUS SOLVER

TABLE S6. Acceptance tests of the steady Navier-Stokes solver. All pass.

Test	What is checked	Result
H1	Poiseuille flow, no slip and Navier slip $S = 0.5$, 2, ∞	exact to 10^{-12}
H2	Brinkman-Poiseuille flow, Da = 10, $S = 0.5$; Brinkman plug flow	velocity error $O(h^{2.7})$, 1.2×10^{-6} at $h = 25$ nm; plug flow exact
H3	Stokes reciprocity of a side-loop valve, three (Da, S)	$D - 1 = 2 \times 10^{-12}$ to 2×10^{-11}; wall flux exactly zero
H4	Kovasznay flow [2], Re = 40	velocity error $O(h^{2.9})$, 2.2×10^{-5} at $h = 25$ nm
H5	backward-facing step [3, 4], expansion ratio 2	reattachment at 2.90 step heights (Re = 100) and 4.96 (Re = 200)
H6	side-loop valve as a classical no-slip device	$D = 0.9993$, 1.0037, 1.0197 at Re = 25, 50, 100
H7	mesh convergence, side-loop valve, Re = 10	$D - 1 = 1.01$, 1.10, 1.14×10^{-2} at wall spacing 20, 14, 10 nm
H8	mirror-symmetric constriction, Re = 10	$D - 1 = 7 \times 10^{-6}$
H9	$(D-1)$/Re at Re = 0.05 to 0.2; mirrored and re-meshed device	constant to 2 %; directional solver bias 6×10^{-7}
U1	units	Poiseuille and Brinkman resistances equal the Gurzhi and Drude formulas to 10^{-12}

Below Re ≈ 1 the diodicity is under the discretisation floor of about 3×10^{-5} and is reported as a bound. For the classical valve with no-slip walls the steady branch continued from Stokes flow ends at a fold near Re = 85; this is

outside the range relevant to graphene.

TABLE S7. Mesh convergence of the classical Tesla valve (175 nm channel, no damping): $D - 1$ at wall spacings of 20 nm (16 703 nodes, the production mesh), 14 nm (25 447 nodes) and 10 nm (38 449 nodes).

Wall condition	Re	20 nm	14 nm	10 nm
perfect slip	3	0.0119	0.0118	0.0124
perfect slip	10	0.234	0.232	0.243
perfect slip	30	1.472	1.510	1.619
slip length $0.5\,W$	3	0.0068	0.0068	0.0072
slip length $0.5\,W$	10	0.151	0.151	0.159
slip length $0.5\,W$	30	1.145	1.177	1.257

Table S7 shows that the diodicity of the classical valve changes by less than 3% between the two coarser meshes and by 5 to 7% between the two finer ones; the pressure drop itself converges only at first order because of the re-entrant corners at the loop junctions. The values quoted in the main text are from the 20 nm mesh and are good to about 10%.

S4. PLASMONIC SOLVER

TABLE S8. Acceptance tests of the perturbation solver (P1 to P9, P15 to P17) and of the time-domain solver (P10 to P14). All pass. In the code the steady gated tests P18 to P20 below are numbered T18 to T20.

Test	What is checked	Result
P1	static limit against the viscous solver	relative difference 5×10^{-4}
P2	linear plasmon response of a channel against $1/\cos kL$	errors 7×10^{-9} to 1.2×10^{-5}
P3	position and width of the first plasmon resonance, with friction, viscosity and viscoelastic viscosity	position 0.1568 (exact 0.1571); widths within 1 %
P4	photoresponse against the Dyakonov-Shur closed form [5] and an independent quadrature	errors 2×10^{-7} to 2×10^{-4}
P5	funnel, quasi-static rectified current: curvature of the steady solution against perturbation	2.206×10^{-2} against 2.210×10^{-2}
P6	time domain against perturbation at small amplitude	ratios 0.9997 and 0.9994
P7	symmetric termination: dc current of a symmetric constriction and of a channel relative to the funnel	6×10^{-7} and 10^{-8} against 1.6×10^{-3}
P8	mesh convergence of the funnel response on three meshes	last change 2.0 % and 1.1 % at reduced frequency 1 and 3
P9	units	plasma speed, gate overdrive, resistance and frequency units against hand values
P10	time-domain efficiency against the time-discrete perturbation law	ratios 1.0033 and 1.0038
P11	dc load correction from a deliberately wrong bias seed	1.0016 (short circuit), 1.0017 (open circuit)
P12	time step: 60 against 120 steps per period at 100 K	efficiency ratio 0.991
P13	power bookkeeping	absorbed power within 0.2 % of the linear value
P14	P10 for the symmetric termination	ratio 0.998
P15	contact resistance: $I_{\mathrm{sc}}(R_{\mathrm{dc}} + 2R_c) = V_{\mathrm{oc}}$; port relations; passivity	7×10^{-15}; 4×10^{-16}; absorbed fraction 0.39
P16	invariance under co-scaling of density, gate distance, width and scattering times at fixed nondimensional groups	nondimensional response identical to round-off
P17	units and closure at a second operating point	closed forms to 2×10^{-16}; Dirac against shallow-water response ratio 0.970

The time-domain runs start from the perturbative state of the time-discretised operator, which removes a start-up transient that would otherwise exceed the rectified signal at small amplitude; the dc load current is held at the small-signal prediction and corrected with the dc Thevenin relation. The level of the square law carries the 60-step time-discretisation bias (+0.6%, +4.5% and −0.5% at 0.1, 0.3 and 1 THz) and the 2% mesh uncertainty of P8; the ratios to the square law quoted in the main text are taken against the time-discrete law, so the bias cancels. The

shape of the response is converged in the time step: at 100 K the efficiency ratio between 60 and 120 steps per period is 0.991 (P12), and at 300 K the ratio of the efficiency to the square law changes from 0.992, 1.054, 1.101 to 1.004, 1.058, 1.109 at amplitudes 0.1, 0.5, 0.7 of the voltage unit at 0.3 THz, and from 1.144 to 1.145 at 0.7 at 0.1 THz.

TABLE S9. Acceptance tests of the steady gated solver, with the gate overdrive U_0 as the unit of bias. All pass.

Test	What is checked	Result
P18	uniform gated channel, steady voltage against the one-dimensional closed form with inertia, both closures, both signs, up to $0.3\,U_0$	worst relative error 1.1×10^{-4}
P19	mirror-symmetric constriction at $0.1\,U_0$: asymmetry under exchange of the contacts against asymmetry under reversal of polarity	3.8×10^{-6} against 4.96×10^{-2}
P20	incompressible limit: funnel, slip walls, Re = 10, Mach number 10^{-3}, against the viscous solver on the same mesh	$D = 0.83845$ against 0.83843

TABLE S10. Mesh convergence of the steady gated flow in the 175 nm funnel (300 K parameters, slip walls): asymmetry under exchange of the contacts at two biases, its linear coefficient a and quadratic coefficient b in $D - 1 = a\,(V/U_0) + b\,(V/U_0)^2$ from those two points, and the asymmetry under reversal of polarity at $V/U_0 = 0.03$, for which the closed form is 1.4521×10^{-2}.

Wall spacing	exchange, 0.01	exchange, 0.03	a	b	polarity, 0.03
28 nm	-1.0×10^{-4}	-3.3×10^{-4}	-9.7×10^{-3}	-0.044	1.4246×10^{-2}
20 nm	-4.0×10^{-5}	-1.5×10^{-4}	-3.5×10^{-3}	-0.049	1.4434×10^{-2}
14 nm	-1.9×10^{-5}	-8.9×10^{-5}	-1.4×10^{-3}	-0.051	1.4498×10^{-2}
10 nm	-7.4×10^{-6}	-5.4×10^{-5}	-2.1×10^{-4}	-0.053	1.4535×10^{-2}

The linear coefficient a of Table S10 is forbidden by the one-dimensional argument of the main text and falls with the mesh size, while the quadratic, inertial coefficient b is stable; the low-bias exchange asymmetry is therefore reported as a bound. For the Tesla valve the exchange asymmetry at $V/U_0 = 0.01$ falls from 2×10^{-6} to 1.5×10^{-7} between wall spacings of 20 and 10 nm. The mirror-symmetric constriction of P19 has an unstructured mesh that is not itself mirror symmetric, so its exchange asymmetry measures the discretisation floor. One production point did not converge, the straight channel at 100 K and $0.3\,U_0$, where the drift speed would exceed the Fermi velocity several times over; it is not used.

S5. PRODUCTION PARAMETERS

Ballistic dc: 400 000 trajectories per bias point in 64 batches. Ballistic ac: 10^6 trajectories per contact and point (4×10^6 at $\lambda = 700$ nm and at the smaller amplitude), chords limited to 40 nm, to $v_F/40f$ and to one twentieth of the field region. Length scaling: static curves at ± 0.05 and ± 0.1 V with 2×10^6 trajectories per side (4×10^6 with scattering), frequency sweeps from 10 GHz to 30 THz with 10^6 (4×10^6) per side, stopped where fL/v_F exceeds about 20; 174 sweep points, 22 controls. Computed field: grid spacing 5.013 nm, chosen so that both contact planes lie on node columns, time step 2.5 fs, 8×10^6 particles for the driven runs and 4×10^6 for the static runs and at 0.1 THz; averages over at least 10 ps and six periods (20 periods for the runs of Fig. 10(f)) after a settling time of 3 ps or more; the rectified current is the mean of the currents through the source plane and the drain plane; 66 runs. Steady gated flow: wall spacing 20 nm for the 175 nm devices and 80 nm for the 1 μm devices, seven biases from 10^{-3} to 0.3 of the gate overdrive, four steady solutions per point (two polarities for each of the two gate connections), 112 points. Viscous: Taylor-Hood meshes with 20 nm wall spacing (23 000 to 55 000 triangles), Newton residual below 10^{-10}. Plasmonic small signal: wall spacing 14 nm for the 175 nm devices and 80 nm (18 601 nodes) for the 1 μm funnel, 25 frequencies from 10 GHz to 3 THz. Plasmonic large signal: the same 1 μm mesh, 60 steps per period, eight amplitudes from 0.01 to 0.7 of the voltage unit.

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